

TIEHO LARA

SKILLS

Qualification: BSc. Geomatics – Land Surveying, *University of Cape Town – South Africa*

Experience: 5 Years

- **Surveying & Geodesy:** Control surveys, projections, datum concepts, spatial reference systems
- **GIS:** Geospatial analytics using desktop applications, and web-based platforms.
- **Remote Sensing:** UAV photogrammetry, aerial imagery processing
- **Database management:** Spatial databases, developer libraries, Statistical analytics, Data Manipulation, Metadata, and Quality assurance
- **Platforms:** Local, Web and Cloud-based GIS systems

Qualification: Ndip. Civil Engineering, *Lerotholi Polytechnic – Lesotho*

Experience: 11 Years

- **Land Survey** | Engineering and Cadastral Survey |
- **Structural Designs and Detailing** | Steel & Timber Design – Prokon, SABS Design Sheet, CAD |
- **Hydrology** | Water Reticulation Systems – Utility Mapping | Ground Water Dynamics | Water Modelling and Prediction |
- **Highway Engineering** | Geometric Designs for Roads | Maintenance Strategies | Drainage Systems | EIA |
- **Geotechnical Engineering** | Soil Investigations & Laboratory Tests | Geological Hazards | Foundations & Retaining walls Designs |

WORK HISTORY

Giri Consultants T/A Giri Group (PTY) Ltd

Land Surveyor – Land Survey, Remote Sensing, & GIS Solutions

May 2022 – Present | Lesotho

- Conducted topographic, cadastral, photogrammetric, and engineering surveys for private and business clients.
- Produced topographic maps, cadastral plans, orthomosaics, and other survey deliverables.
- Established and verified survey control networks, benchmarks, and reference points.
- Collected, processed, and analysed survey data using GNSS, total station, and levelling instruments.



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WEBSITES, PORTFOLIOS, PROFILES

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GitHub: <https://github.com/LaraTieho>

CAREER ACHIEVEMENTS

Organization: GDI Advisors

Summary Impact:

- Established a **complete national GIS dataset** for health catchment analysis.
- Delivered **first-draft catchment maps for 8 districts**, marking a major milestone toward national coverage.
- Shifted project maturity from **data acquisition to analytics, reporting, and decision support**.
- Positioned GIS outputs for **facility-level validation, MoH adoption, and RED-compliant microplanning**.

Next Milestones:

- Complete facility-level validation for all 8 districts.
- Finalize national catchment maps and analytics dashboards.
- Publish standardized reports and WebGIS products for MoH use.
- Institutionalize update workflows for sustainability.

Organization: World Vision Lesotho

- Led the **development of the national MHEWED web system**, working with stakeholders from Lesotho Meteorological Services improving early warning (EW) and early detection (ED) accuracy by **30%**
- Managed 3 drone pilots for collection of Aerial Images data (orthophotos), enhancing real-time disaster monitoring and response.

- Prepared survey drawings and maps in CAD/GIS environments and ensured compliance with technical standards.
- Performed boundary re-establishment, staking, as-built surveys, and field quality checks.

GDI Advisors

Senior GIS Consultant – Spatial Systems & Health Catchments

April 2025 – Present | Lesotho

- Lead **district and national-level geospatial pre-analysis**, contributing to standardized spatial frameworks aligned with government systems.
- Design and implement **end-to-end GIS data pipelines**, including data structuring, validation, metadata assignment, and quality assurance.
- Support **authoritative dataset development**, including facility and village-level spatial validation relevant to NSDI foundations.
- Develop **WebGIS platforms** that enable interoperability, version control, and long-term system sustainability.
- Facilitate technical workshops and capacity building for government stakeholders on spatial data validation and governance.

Clinton Health Access Initiative (CHAI)

GIS & Immunization M&E Analyst

August 2023 – April 2025 | Lesotho

- Supported the Ministry of Health through **spatial data analysis and system strengthening**, contributing to standardized national datasets.
- Applied GIS for **equity, access, and service coverage analysis**, ensuring spatial consistency across administrative levels.
- Strengthened institutional capacity in **spatial data use, quality assurance, and reporting**.
- Ensured alignment of GIS outputs with HMIS and national data standards.

World Vision

GIS Analyst – Remote Sensing & Disaster Risk Reduction

January 2022 – August 2023 | Lesotho

- **Led aerial photogrammetry and UAV-based mapping** for disaster risk reduction initiatives.
- Managed drone operations, including data acquisition and integration into GIS systems.
- Produced decision-support maps using standardized spatial workflows.
- Supported community-level planning through spatial analysis and mapping.

10 Talents Design Build Firm

Civil & Structural Engineering Consultant

August 2015 – December 2021 | Lesotho

- Processed 50+ aerial photogrammetry datasets, creating hazards and floods modelling maps, aiding in 20% faster emergency response.
- Developed GIS-driven reports that influenced **ECHO Phase IV** funding for disaster management and risk.
- Designed 150+ mapping strategies aligned with Community Disaster Preparedness Plans (CDPPs), improving community resilience in 4 districts.
- Assessed 10+ M&E reports, refining mapping accuracy and increasing monitoring efficiency by 30%.

Organization: Clinton Health Access Initiative

- Developed 10+ spatial maps identifying immunization gaps, leading to targeted interventions in under-served areas. Utilized GIS to analyze coverage trends across 10 districts, improving HPV vaccination coverage to 93% for girls of age 9 to 14 years
- Designed an M&E framework for data clerks covering 50,000+ beneficiaries, tracking coverage and equity. Led 3 program evaluations, optimizing vaccination data records in DHIS2. Increased data-driven decision-making by 40% through improved reporting systems.
- Managed a database of 100,000+ immunization records in DHIS2, ensuring 98% data accuracy. Developed real-time dashboards that improved reporting efficiency by 30% during HPV campaigns. Automated data collection processes, reducing manual errors by 25%.
- Trained 50+ data clerks, based at Health Centres, on GIS and M&E tools, enhancing their analytical skills. Partnered with 4+ international organizations to integrate GIS in health planning. Strengthened community engagement, leading to a 15% increase in immunization uptake.
- Conducted routine data audits, improving data integrity by 95%. Standardized data validation processes, ensuring compliance with WHO and national health data standards. Enhanced reporting accuracy

- Applied engineering surveying, CAD design, and spatial reasoning in infrastructure projects.
- Conducted site inspections and supported regulatory compliance.
- This engineering foundation strengthens applied geodesy, surveying accuracy, and GIS integration.

through rigorous quality checks, reducing discrepancies by 20%.

- Assessed 10+ M&E reports, refining mapping accuracy and increasing monitoring efficiency by 30%. making informed decisions that drive success

REFERENCES

Organization: GDI Advisors

Title: GIS Lesotho Regional Coordinator

Email Address: Abdoulaye.dia@gdi-advisors.com

Contacts: +33782353130

Organization: Clinton Health Access Initiative - Lesotho

Title: CHAI Country Director

Email Address: emandara@clintonhealthaccess.org

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Organization: World Vision - Lesotho

Title: ECHO Project Manager

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PROJECT MANAGEMENT AND COLLABORATION TECHNOLOGY SKILLS

- Asta Powerproject
- Microsoft Projects
- Microsoft 365 suite
- Confluence platform
- Notion platform